Proofs for the dog game with inertia

This note proves the two propositions of Section 5.20 of the research report. The proofs are mine and have not been refereed; every step is also checked numerically in `tests/test_inertia_theory.py` and the studies `inertia_explicit` and `inertia_random`.

Setting

Rewards are sums over the two coordinates, the dog moves coordinate by coordinate, and the action set is the product of two intervals. So a profile is a feedback equilibrium of the two-dimensional game exactly when it is one in each coordinate: a two-dimensional deviation is a pair of one-coordinate deviations whose payoffs add. Work in one coordinate from now on.

The state is the dog's position x in [0, 1]. Red and blue choose actions a_red and a_blue in [0, 1], and the next position is

f(x, a_red, a_blue) = w0 x + (1 - w0) (w a_red + (1 - w) a_blue),

with inertia w0 in [0, 1), red's weight w in (0, 1) and blue's weight 1 - w. Player i receives -(f - h_i)^2, the squared distance of the next position from its house h_i, discounted by g in [0, 1). A stationary Markov profile pi maps positions to actions, and V_i(x) is player i's discounted payoff from x under pi. The profile is a feedback (Markov-perfect) equilibrium if, at every x, each player's action maximizes -(f - h_i)^2 + g V_i(f) given the other's action. By the one-shot deviation principle (Fudenberg and Tirole, 1991), which applies because payoffs are bounded and discounted, this is the same as having no profitable one-round deviation at any state.

Let (a_red*, a_blue*) be the one-shot Nash actions and d* = w a_red* + (1 - w) a_blue* the one-shot dog. Define

k = (1 - g w0^2) / (1 - g w0), T_i = d* + k (h_i - d*), z_h(x) = w0 x + (1 - w0) d*.

k is at least 1 because w0^2 <= w0. T_i is player i's target and z_h(x) is the next position when both hold.

Lemma 1 (the value of holding)

If both players always play the one-shot actions, then z_h(x) - d* = w0 (x - d*), so the position after t + 1 rounds is d* + w0^(t+1) (x - d*), and summing the geometric series gives, with u = x - d* and e = d* - h_i,

V_i(x) = -[e^2 / (1 - g) + 2 e w0 u / (1 - g w0) + w0^2 u^2 / (1 - g w0^2)].

Lemma 2 (the one-round objective under holding)

If player i plays something else for one round and both hold afterwards, its payoff as a function of the next position z is G_i(z) = -(z - h_i)^2 + g V_i(z). Its second derivative is -2 - 2 g w0^2 / (1 - g w0^2) < 0, so it is strictly concave, and

G_i'(z) = -2 [(z - d*) / (1 - g w0^2) + (d* - h_i) / (1 - g w0)],

which is zero exactly at z = T_i. So G_i increases up to T_i and decreases after it. With the other player at its one-shot action, player i reaches the next positions in an interval of length (1 - w0) w_i (w_red = w, w_blue = 1 - w), which is increasing in its action, and holding gives z_h(x).

Proposition 1

Holding the one-shot actions at every position is a feedback equilibrium if and only if each player at the top of the box ($a_i^* = 1$) has $k(h_i - d^*) \geq w_0(1 - d^*)$, each player at the bottom ($a_i^* = 0$) has $k(d^* - h_i) \geq w_0 d^*$, and, when $w_0 > 0$, no one-shot action is strictly inside the box.

Proof. By Lemma 2 and the one-shot deviation principle, holding is optimal for player i at x exactly when $z_h(x)$ maximizes the strictly concave G_i over i's reachable interval. For a player at the top, $z_h(x)$ is the right end of that interval, so it is optimal iff $G_i'(z_h(x)) \geq 0$, that is $z_h(x) \leq T_i$, that is

$$w_0(x - d^*) \leq k(h_i - d^*).$$

The left side is largest at $x = 1$, which gives the stated condition (for $w_0 = 0$ it always holds, since a one-shot best response at the top has $h_i \geq d^*$). A player at the bottom is the mirror image, with the worst case at $x = 0$. For a player strictly inside, $z_h(x)$ is inside its interval, so holding is optimal iff $z_h(x) = T_i$. A one-shot best response strictly inside the box puts the dog on the player's house, $h_i = d^*$, so $T_i = d^*$, and the condition becomes $w_0(x - d^*) = 0$, which fails at every x other than d^* when $w_0 > 0$. Optimality at every x for every player is the stated set of conditions. ■

Dividing by $1 - d^*$ (or by d^*), the conditions say that the player's slack $s = (h_i - d^*)/(1 - d^*)$ (or $(d^* - h_i)/d^*$) is at least $\rho(w_0, g) = w_0(1 - g w_0)/(1 - g w_0^2)$. ρ rises from 0 to 1 as w_0 goes from 0 to 1 (its derivative in w_0 is $(1 - g w_0(2 - w_0))/(1 - g w_0^2)^2$, at least $(1 - g)/(1 - g w_0^2)^2 > 0$), and its derivative in g is $w_0(w_0^2 - w_0)/(1 - g w_0^2)^2 \leq 0$, so more patient players tolerate more inertia.

Proposition 2

Suppose the coordinate is straddling: after relabeling, $h_{blue} \leq w \leq h_{red}$, so the one-shot actions are $a_{red}^* = 1$, $a_{blue}^* = 0$ and $d^* = w$. Then the policy

$$a_{red}(x) = \text{clip}(1 + (T_{red} - z_h(x))/((1 - w_0) w), 0, 1), \quad a_{blue}(x) = \text{clip}(0 + (T_{blue} - z_h(x))/((1 - w_0)(1 - w)), 0, 1)$$

is a feedback equilibrium for every w_0 in $[0, 1)$ and g in $[0, 1)$.

In words, each player aims the next position at its target, as far as its weight allows, as if the other held its one-shot action.

Proof. If $w_0 = 0$ then $z_h = d^*$, both clips return the one-shot actions, and the game is the one-shot game in every round, so assume $w_0 > 0$. Note $T_{blue} \leq d^* \leq T_{red}$, because $k \geq 1$ and $h_{blue} \leq d^* \leq h_{red}$.

Regions. Let $X_{red} = d^* + (T_{red} - d^*)/w_0$ and $X_{blue} = d^* - (d^* - T_{blue})/w_0$, so that $z_h(X_{red}) = T_{red}$ and $z_h(X_{blue}) = T_{blue}$. Red eases off its edge exactly on $R = \{x : z_h(x) > T_{red}\} = (X_{red}, 1]$, blue exactly on $B = [0, X_{blue})$, and both hold on $H = [X_{blue}, X_{red}]$ intersected with $[0, 1]$. R and B do not meet, because $X_{blue} \leq d^* \leq X_{red}$, so at most one player leaves its edge at any x. The next position under the policy is

$$N(x) = z_h(x) \text{ on } H, \quad \max(T_{red}, z_h(x) - (1 - w_0) w) \text{ on } R, \quad \min(T_{blue}, z_h(x) + (1 - w_0)(1 - w)) \text{ on } B.$$

N is nondecreasing; on R, $T_{red} \leq N(x) < x$ (since $x > X_{red} \geq T_{red}$ and $z_h(x) < x$ when $x > d^*$); on B, $x < N(x) \leq T_{blue}$. H is invariant: for x in H, $N(x) = z_h(x)$ lies between x and d^*, both in H. The targets lie in H when they lie in $[0, 1]$: $T_{red} \leq X_{red}$ because $T_{red} - d^* \geq 0$ and $w_0 \leq 1$, and $T_{red} \geq d^* \geq X_{blue}$; similarly for T_{blue}.

Values. Because H is invariant and both hold on it, V_i equals the holding value of Lemma 1 on H. Let $g_i(z) = -(z - h_i)^2 + g V_i(z)$ be player i's one-round objective as a function of the next position. On H it is G_i of Lemma 2. At any z the policy moves the dog to $N(z)$, so $V_i(z) = g_i(N(z))$, and therefore

g_i(z) = -(z - h_i)^2 + g g_i(N(z)) for every z. (*)

Claim: g_red is nondecreasing on [0, T_red] and nonincreasing on [T_red, 1] (each part when nonempty).

Above the target. Let M be the set of bounded functions phi on [T_red, 1] that equal G_red on [T_red, X_red] and are nonincreasing. Define (A phi)(z) = G_red(z) on [T_red, X_red] and (A phi)(z) = -(z - h_red)^2 + g phi(N(z)) on R; this is well defined because N maps R into [T_red, 1). A is a contraction with modulus g in the sup norm, and by (*) the restriction of g_red to [T_red, 1] is its fixed point. A maps M into M:

- on R, -(z - h_red)^2 decreases, because z > X_red >= T_red >= h_red (T_red - h_red = (k - 1)(h_red - d*) >= 0), and phi(N(z)) does not increase, because N does not decrease and phi does not increase;
- across the boundary, for z1 in [T_red, X_red] and z2 in R, (A phi)(z2) <= -(z2 - h_red)^2 + g phi(T_red) < -(X_red - h_red)^2 + g G_red(T_red) = G_red(X_red) <= G_red(z1), using N(z2) >= T_red, phi(T_red) = G_red(T_red), the identity G_red(X_red) = -(X_red - h_red)^2 + g G_red(z_h(X_red)) with z_h(X_red) = T_red (this is (*) on H), and that G_red decreases after T_red.

M is closed under uniform limits and contains, for example, G_red on [T_red, X_red] extended by the constant G_red(X_red), so iterating A from it converges to the fixed point, which therefore lies in M (Banach's fixed point theorem).

Below the target. The same argument on [0, T_red] with the set of functions that equal G_red on [X_blue, T_red] and are nondecreasing: on B, -(z - h_red)^2 increases because z < X_blue <= d* <= h_red, and N maps B into (z, T_blue], which lies in [0, T_red]; across the boundary, (A phi)(z1) <= -(z1 - h_red)^2 + g G_red(T_blue) < -(X_blue - h_red)^2 + g G_red(T_blue) = G_red(X_blue) <= G_red(z2) for z1 in B and z2 in [X_blue, T_red], using z_h(X_blue) = T_blue, T_blue in [X_blue, T_red] and that G_red increases up to T_red. This proves the claim; the claim for blue is the mirror image, with its peak at T_blue.

Best replies. A function that does not decrease up to T and does not increase after it is maximized over any interval [L, U] at clip(T, L, U). Given blue's action at x, red reaches the next positions [U - (1 - w0) w, U], where U is the position with a_red = 1. On H and R, blue is at 0, so U = z_h(x) and red's best reply is clip(T_red, U - (1 - w0) w, U), which is exactly the policy (it holds on H, where z_h(x) <= T_red, and moves to max(T_red, z_h(x) - (1 - w0) w) on R). On B, blue moves the position up to N(x) <= T_blue <= T_red with red at 1, so U = N(x) <= T_red and the best reply is red's edge, as in the policy. Blue's case is the mirror image. By the one-shot deviation principle the policy is a feedback equilibrium. ■

Many houses

With N players, weights w_i summing to 1 and the dog at sum_i w_i a_i, Lemmas 1 and 2 hold unchanged for each player (the other players' held actions only enter through d*, and player i reaches an interval of length (1 - w0) w_i). So Proposition 1 holds verbatim: holding the one-shot actions is a feedback equilibrium iff every player at an edge has slack at least rho(w0, g) and, when w0 > 0, no player (the pivot) is strictly inside the box. In the one-shot equilibrium, which is the generalized median of Renault and Trannoy (2005), at most one player is inside. The study `many_houses` measures how often there is none, and how the largest admissible inertia scales with N. The target form also carries over numerically: if every player aims at a target, each round is the one-shot game with the targets as houses, and targets from a fixed point give a feedback equilibrium whenever every player's one-round objective is single-peaked (study `many_targets`); there is no proof for many houses.

Continuous-time limit

With time step dt, w0 = 1 - dt and g = exp(-r dt), 1 - g w0 = (r + 1) dt + O(dt^2) and 1 - g w0^2 = (r + 2) dt + O(dt^2), so k -> (r + 2) / (r + 1) and rho -> (r + 1) / (r + 2). In Proposition 2, (T_i - z_h(x)) / ((1 - w0) w_i) = (T_i - x + dt (x - d*)) / (dt w_i), which tends to +infinity where x < T_i and to -infinity where x > T_i, so the policy tends to the bang-bang rule "push the edge toward T_i", with the interior band of width O(dt). This describes the limit of the discrete-time equilibria; that it is an equilibrium of the continuous-time game is not proved here.

Overshooting

In Proposition 2 a player aims the next dog position at T_i = d* + k (h_i - d*), so it aims past its own house by (k - 1)(h_i - d*). Since k = (1 - g w0^2) / (1 - g w0), dk/dw0 has the sign of g w0^2 - 2 w0 + 1, which vanishes at w0 = (1 - s) / g = 1 / (1 + s) with s = sqrt(1 - g). So k = 1 at w0 = 0 and at w0 = 1, and the overshoot is largest at the intermediate inertia 1 / (1 + s), where k = 2 / (1 + s): at most 2, approached by patient players.

Persistence and the median rule

Now N players (N odd) with distinct houses h_1 < ... < h_N report points r_i in [0, 1], and the outcome moves to z = w0 x + (1 - w0) median(r); losses are (z - h_i)^2, discounted. Let h_m be the median house.

Proposition 3. For w0 = 0 truthful reporting (r_i = h_i at every outcome) is a feedback equilibrium; for any w0 in (0, 1) it is not.

Proof. With w0 = 0 the round does not depend on the outcome, so each round is the one-shot median game, where truthful reporting is dominant (Moulin, 1980). With w0 > 0 and everyone truthful, the outcome moves to w0 x + (1 - w0) h_m and converges to h_m, so by Lemma 2 (with d* = h_m, the median player's house) the median player's one-round objective is a concave quadratic in the next outcome peaked at T_m = h_m. By changing its report within [h_(m-1), h_(m+1)] the median player moves the next outcome over an interval that contains w0 x + (1 - w0) h_m strictly inside, and that contains h_m itself whenever x is close enough to h_m. At such an x other than h_m, reporting (h_m - w0 x) / (1 - w0), not h_m, puts the outcome on its peak, a strict gain. ■

Proposition 4. For every w0 in [0, 1) the following profile is a feedback equilibrium: the median player reports clip((h_m - w0 x) / (1 - w0), 0, 1), players with houses above h_m report 1 and players below report 0. Under it the outcome moves to G(x) = clip(h_m, w0 x, w0 x + 1 - w0) each round: as close to the median house as can be reached, the path a planner minimising (z - h_m)^2 would choose.

Proof. The median of the reports is the median player's report, which gives G. By the one-shot deviation principle it suffices to check one-round deviations.

- The median player can move the median, hence the next outcome, anywhere in [w0 x, w0 x + 1 - w0]. Its problem is then a one-player control problem with loss (z - h_m)^2: if h_m is reachable it stays reachable forever after, and otherwise the reachable interval moves monotonically with the outcome, so moving as close to h_m as possible each round is optimal.
- A player with house h_i > h_m can only lower the median (by reporting below the median player's report), never raise it. If G(x) <= h_m, lowering the next outcome brings it further from h_i and, since G is nondecreasing, lowers the whole future path, which stays at or below h_m < h_i: a loss. If G(x) > h_m, the median player is reporting 0, so G(x) = w0 x is already the lowest reachable outcome and nothing can be lowered.
- Players below the median are the mirror image. ■

So inertia costs the median rule its truthfulness but not its speed: the outcome still moves to the median house as fast as any mechanism could move it there. The mean rule has no such property; in the study `mechanisms` its outcome moves more slowly, because players on the far side of the outcome keep pulling against it.

Cooling off

Add a shock to each round: the outcome lands at y = clip(z + e, 0, 1), where z is the outcome the reports aim at and e has a symmetric distribution with e != 0 with positive probability. The shock either comes with the new aggregate, e = (1 - w0) s eps (the inertia filters it), or hits the outcome itself, e = s eps. The greedy planner aims at the mean house hbar of the N houses each round, as close as it can reach.

Proposition 5. Let wbar = min(hbar, 1 - hbar). For w0 in [0, wbar] the greedy planner's expected discounted summed loss, from any start, is

L(w0) = (sum_i (h_i - hbar)^2 + N phi(sigma)) / (1 - g), phi(sigma) = E[(clip(hbar + sigma eps, 0, 1) - hbar)^2],

with sigma = (1 - w0) s when the shock comes with the aggregate and sigma = s when it hits the outcome. So with a filtered shock L is strictly decreasing on [0, wbar]: some inertia is strictly better than none, and the welfare-maximizing inertia is at least wbar. With a shock on the outcome, L is constant on [0, wbar].

Proof. From any y in [0, 1] the reachable interval [w0 y, w0 y + 1 - w0] contains hbar when w0 <= hbar and hbar <= 1 - w0, so the planner aims at hbar in every round and every round's outcome is clip(hbar + e, 0, 1). The summed loss at an outcome y is sum_i (h_i - hbar)^2 + N (y - hbar)^2, which gives L. Each |clip(hbar + sigma eps, 0, 1) - hbar| is nondecreasing in sigma and strictly increasing where it is not clipped, which has positive probability, so phi is strictly increasing. ■

Above wbar the planner can no longer reach hbar from every outcome, and slow adjustment starts to cost; the study `mechanisms_noise` finds the loss U-shaped in w0, lowest at w0 = 0.6 to 0.75 for the planner and at 0.45 to 0.9 for the equilibria of the mean and the median rule (which the proposition does not cover), the optimum rising with the size of the shock. With a shock on the outcome the planner is best off with no inertia. (The greedy planner is not optimal when clipping is frequent; the study also computes the optimal planner.)

What the propositions do not cover

- **Interior one-shot actions.** If a player's one-shot action is strictly inside the box, its hold region is the single point d*, and off it that player pushes at an edge; the dog then returns faster than under holding, the continuation in Lemma 2 is wrong, and the other player's best target moves. In one random game (houses 0.754 and 0.538, w = 0.347, w0 = 0.767, g = 0.335) the dynamic program's red aims at about 0.785 where the policy of Proposition 2 aims at 0.771. The policy fails in 53 of 123 such random games. Because a player's one-round objective depends only on the next position, an equilibrium in target form still exists whenever the objectives stay single-peaked; with the targets from a two-dimensional fixed point (`target_equilibrium`), this holds numerically in 122 of the 123 games. A proof is open.
- **Uniqueness.** The propositions give one feedback equilibrium; they do not rule out others.
- **Shared coordinates.** When the houses coincide, both one-shot actions can be interior and the one-shot game has a continuum of equilibria.